



04 — Deployment Plan

how to ship NexFlow to production. Two paths are described — Replit Deployments (recommended default for speed) and AWS me-south-1 (Bahrain) for GCC customers with hard data-residency requirements. Includes environments, DB strategy, secrets, domains/TLS, observability, backups, cost, and a release checklist.





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Document scope: how to ship NexFlow to production. Two paths are described — Replit Deployments (recommended default for speed) and AWS me-south-1 (Bahrain) for GCC customers with hard data-residency requirements. Includes environments, DB strategy, secrets, domains/TLS, observability, backups, cost, and a release checklist.

1. Environments

ENV	PURPOSE	TARGET AUDIENCE	BRANCH / SOURCE
dev	Day-to-day engineering. Auto-seeded sample data.	Engineers, designers.	The active Replit workspace.
staging	Pre-prod customer demos, regression testing, integration acceptance. Production-shaped data, sanitized.	Solutions engineering, customer pilots.	Mirror of <code>main</code> deployed to a separate Replit Deployment / AWS account.
prod	Live, paying customers.	Customers.	Tagged release from <code>main</code> .

For each env we run:

SERVICE	DEV	STAGING	PROD
Web (<code>artifacts/nexflow</code>)	Vite dev server	Static build behind Replit proxy	Static build behind CDN
API (<code>artifacts/api-server</code>)	Express on <code>\$PORT</code>	Always-on Replit deployment	Reserved-VM Replit deployment / ECS service
Mobile (<code>artifacts/mobile</code>)	Expo dev	EAS preview channel	EAS production channel + App Store / Play Store
Marketing demo	Vite dev	Static build	Static build
DB	Replit Postgres	Replit Postgres (separate)	Replit Postgres (prod) or AWS RDS Postgres me-south-1

2. Recommended path A — Replit Deployments

Best for first 12 months. Lowest operational overhead, fastest time-to-customer.



2.1 What gets deployed

Each artifact is published as its own deployment behind the workspace shared proxy, routed by path (see `docs/02-tech-stack.md §11`):

- `/` → web app (static)
- `/api` → API server (autoscale)
- `/marketing-demo-video/` → demo video (static)

Mobile is shipped via **EAS Build → App Store / Play Store**, not Replit Deployments.

2.2 Deployment types per artifact

ARTIFACT	TYPE	WHY
Web (<code>artifacts/nexflow</code>)	Static	It's a Vite SPA — pure assets. CDN-cached, near-zero ongoing cost.
API (<code>artifacts/api-server</code>)	Autoscale	Express app; bursts during the workday. Autoscale keeps costs flat outside business hours. Set min instances ≥ 1 for paying customers to avoid cold starts.
Marketing demo	Static	Built bundle.
Mobile	n/a (EAS)	Native app stores.

2.3 Pre-deploy checklist

1. Confirm catalog versions are pinned in `pnpm-workspace.yaml`.
2. Run `pnpm run typecheck` at the repo root — must pass.
3. Run `pnpm --filter @workspace/nexflow run build` and `pnpm --filter @workspace/api-server run build` locally.
4. Apply DB migrations against the target environment (see §4).
5. Set all required secrets via the workspace secrets UI (see §5).
6. Ensure `autoSeed` is **disabled** for prod (gate it with `NODE_ENV !== "production"` or feature flag) — sample data is for dev only.

2.4 Domains & TLS

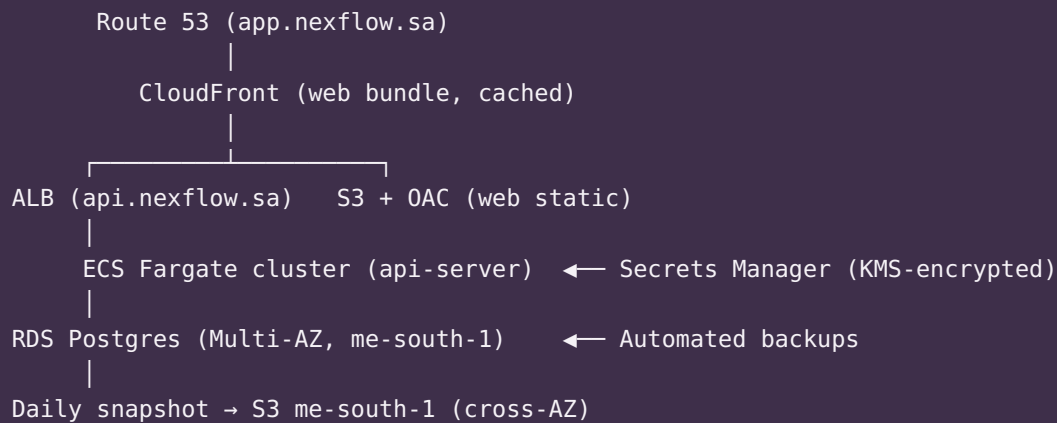
- Replit issues HTTPS automatically via the proxy domains in `$REPLIT_DOMAINS` (comma-separated).
- Custom domains: bind in the Deployment settings. Replit handles cert issuance and renewal.
- Production recommendation: `app.nexflow.ai` → web; `api.nexflow.ai` → API.



3. Alternate path B — AWS me-south-1 (Bahrain) for GCC residency

Some GCC customers (banks, government, regulated insurers) require data to physically reside in-region. Use this path when the contract demands it.

3.1 Reference architecture



3.2 What changes vs. path A

CONCERN	PATH A (REPLIT)	PATH B (AWS ME-SOUTH-1)
API host	Replit Autoscale	ECS Fargate (2× t-shirt sized tasks) behind ALB
Web host	Replit Static	S3 + CloudFront with Origin Access Control
DB	Replit Postgres	RDS Postgres (db.t4g.medium to start; Multi-AZ)
Secrets	Replit Secrets	AWS Secrets Manager
Logging	Replit logs	CloudWatch Logs + Logs Insights
Region	Global Replit edge	me-south-1 (Bahrain) only
Mobile	EAS	Same — EAS

3.3 IaC

Maintain a small **Terraform** module per env in `infra/aws/` (not yet checked in). One module per: VPC, RDS, ECS cluster, ALB, CloudFront + S3, Route 53. Apply via GitHub Actions. CD pushes container images to ECR; ECS rolling-deploys.

3.4 Migration playbook (Replit → AWS, when a customer triggers it)

1. Snapshot Replit Postgres to a `pg_dump`.
2. Restore into RDS Postgres me-south-1.
3. Build & push the API image to ECR.



4. Apply Terraform.
 5. Cut DNS over.
 6. Disable the Replit deployment (keep the workspace as the dev environment).
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■ 4. Database provisioning & migrations

- Source of truth: `lib/db/src/schema/index.ts` (Drizzle).
 - Migrations: generated with `drizzle-kit generate` to `lib/db/drizzle/`. Apply with `drizzle-kit migrate` (or `push` in dev).
 - The workspace shared `database` skill is the dev-side path; the production DB is provisioned per env (Replit Postgres or RDS).
 - **Migration policy:** every PR that touches `schema/index.ts` must: 1. Generate a new migration file. 2. Be backward-compatible for at least one release (additive columns; no destructive renames in a single deploy). 3. Be smoke-tested against staging before prod.
 - **Auto-seed:** the boot-time `autoSeed()` in `artifacts/api-server/src/lib/autoSeed.ts` populates the "default" org with sample CRM data. **Must be gated off in prod** (e.g. `if (process.env.NODE_ENV === "production") return;`).
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■ 5. Secrets management

All secrets must come from the platform secrets store, never from code.



VARIABLE	WHERE IT'S SET	REQUIRED BY
<code>DATABASE_URL</code>	Replit Secrets / AWS Secrets Manager	api-server, lib/db
<code>AI_INTEGRATIONS_OPENROUTER_BASE_URL</code> / <code>_API_KEY</code>	Replit Connectors (preferred)	api-server
<code>AI_INTEGRATIONS_OPENAI_BASE_URL</code> / <code>_API_KEY</code>	Replit Connectors	api-server
Resend key	Replit <code>resend</code> connector or <code>RESEND_API_KEY</code>	api-server
Twilio Account SID / Auth Token	Set when telephony goes live	api-server (<code>routes/calls.ts</code>)
WhatsApp Business token	Set when WhatsApp goes live	api-server (<code>routes/marketing.ts</code> , <code>messages UI</code>)
Payment provider keys (Mada / Tap / HyperPay / PayTabs)	Set when Quote-to-Cash goes live	api-server
<code>REPL_IDENTITY</code> / <code>WEB_REPL_RENEWAL</code> / <code>REPLIT_CONNECTORS_HOSTNAME</code>	Auto-injected by Replit	api-server

When migrating to AWS, secrets move from Replit Secrets to **AWS Secrets Manager** and are injected into ECS task definitions via `secrets:` blocks (KMS-encrypted at rest, in transit via the ECS agent).

5.1 Auth (planned, not yet wired)

The app currently runs in demo mode (no auth wall). Before the first paying customer:

- **Default:** Replit Auth (OpenID Connect with PKCE). Wire via the workspace `replit-auth` skill — adds auth routes, middleware, web hook, and mobile auth in one pass.
- **Enterprise / SAML / SSO:** Clerk (per the workspace `clerk-auth` skill) — needed if a customer requires SAML or SCIM provisioning.

Scope this work into a dedicated PR before any GA launch.



6. Observability & logging

CONCERN	PATH A (REPLIT)	PATH B (AWS)
App logs	Pino → stdout → Replit log viewer	Pino → stdout → CloudWatch Logs
HTTP request logs	<code>pino-http</code> (already wired in <code>app.ts</code>)	Same
Errors	Inline <code>try/catch</code> + <code>logger.error</code> . AI route <code>try/catch</code> returns empty rather than crashing.	Same. Forward to Sentry (org-wide DSN).
Uptime	Replit health check on <code>/api/healthz</code>	Route 53 health check + CloudWatch alarm
Metrics	Replit deployment metrics	CloudWatch + Container Insights
AI cost	OpenRouter dashboard	Same — OpenRouter is provider-agnostic to the host

Add **Sentry** (`@sentry/node` for the API, `@sentry/react` for the web) before paying-customer launch. One DSN per env.

7. Backup & disaster recovery

TIER	RPO	RTO	MECHANISM
A: Replit Postgres	24 h	4 h	Daily snapshot. Restore via Replit DB UI.
B: RDS Postgres me-south-1	5 min (PITR)	1 h	Automated backups + Multi-AZ failover. Cross-region snapshot copy to <code>eu-west-1</code> weekly for catastrophic-region recovery.

Quarterly DR drill: restore yesterday's snapshot into a scratch DB, run a Drizzle migration, verify a known query.

For object storage (call recordings, business-card images, AI-generated assets), use the workspace `object-storage` skill (S3-compatible). Versioning **on**; lifecycle rule deletes after 90 days unless tagged `legal-hold`.



8. Cost-by-environment estimate (monthly)

Order-of-magnitude only. Re-baseline at every contract.

Path A — Replit Deployments

ITEM	DEV	STAGING	PROD (10 CUSTOMERS, ~50 REPS)
Web (Static)	\$0	\$0	~\$10
API (Autoscale, min 1)	n/a	~\$25	~\$120
Postgres	included	included	~\$25
Object storage	\$0	~\$5	~\$20
OpenRouter (LLM usage)	\$50	\$100	~\$600 (heavy AI org)
Resend	\$0	\$20 (Pro)	\$20-\$80
Subtotal	~\$50	~\$150	~\$800-900

Path B — AWS me-south-1

ITEM	PROD (SINGLE CUSTOMER, ~50 REPS)
ECS Fargate (2 tasks, t-shirt small)	~\$60
ALB	~\$20
RDS Postgres db.t4g.medium Multi-AZ	~\$160
CloudFront + S3 (web)	~\$10
CloudWatch Logs	~\$10
Secrets Manager	~\$5
Backups (snapshots + cross-region)	~\$15
OpenRouter	~\$300-600
Resend	~\$20
Subtotal	~\$600-900

Costs grow linearly with seats; LLM usage is the variable item.



■ 9. Release checklist

For every production push (web + API + DB):

- [] CI green: `pnpm run typecheck` + builds for all artifacts.
- [] DB migrations are additive and reviewed.
- [] `autoSeed` confirmed off in prod (or gated).
- [] Secrets present in target env (DB, OpenRouter, OpenAI, Resend, Twilio when live).
- [] Smoke test on staging:
- [] Sign in (when auth is live).
- [] Open `/`, `/funnel`, `/calls`, `/assistant`.
- [] Send a test campaign email through Resend.
- [] Trigger an AI assistant chat.
- [] Run one workflow.
- [] Sentry release tag created.
- [] Mobile EAS build promoted to the matching channel.
- [] Changelog updated; customer-facing release note drafted.
- [] Roll-back plan: previous Replit deployment kept warm for 24 h; previous container image kept in ECR for 7 days.

■ 10. Routine ops cadence

CADENCE	TASK
Daily	Triage new errors in Sentry. Spot-check OpenRouter spend.
Weekly	Review API p95 latency. Ensure auto-seed didn't run in prod.
Monthly	Rotate Resend / Twilio / payment-provider keys if customer compliance requires it.
Quarterly	DR drill (§7). Cost review. Catalog version refresh.